LECTURE · IP AND APPROPRIABILITY

The protection of innovation

Types of IPR and appropriability

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Monday 5 October 2026

Politecnico di Torino

00

Some useful information (and thoughts) before starting



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IMPORTANT: enroll in the Project Work

MANDATORY

Register in the Project Work enrollment sheet (teams revealed during kick-off)

ENROLLMENT SHEET
docs.google.com/spreadsheets/d/10uMG_BA1tBhXWHaWk4roumaW4o9qDWZh2QbGOXyKCnk

SCAN TO ENROLL

Questions on teams and groups:
Fatima, student tutor, responsible for PW organization:
syedafatima.wasti@studenti.polito.it

My part of the course: calendar

LECTURES · 12 × 90 MIN

MON 05 OCT	**Types of IPR and appropriability** TODAY	13:00–14:30 · 4P
MON 12 OCT	**Patents: filing, history, litigation**	13:00–14:30 · 4P
MON 19 OCT	**IPR as a strategic choice for companies**	13:00–14:30 · 4P
TUE 20 OCT	**Patent intelligence and landscaping**	17:30–19:00 · R3
WED 28 OCT	**Knowledge spillovers**	13:00–14:30 · R3
WED 04 NOV	**Patent valuation**	13:00–14:30 · R3
WED 18 NOV	**IP and AI**	13:00–14:30 · R3
MON 07 DEC	**Digital strategy I**	13:00–14:30 · 4P
WED 09 DEC	**Digital strategy II**	13:00–14:30 · R3
MON 14 DEC	**Digital strategy III**	13:00–14:30 · 4P
TUE 15 DEC	**AI case study**	17:30–19:00 · R3
WED 16 DEC	**Integrating strategy in landscape**	13:00–14:30 · R3

PROJECT WORK SESSIONS
8 x 90 min

TUE 06 OCT	**Kick-off** 17:30–19:00 · R3
TUE 13 OCT	**Data collection: TIP platform** 17:30–19:00 · R3
WED 21 OCT	**Tutorial 1: patent attributes** 13:00–14:30 · R3
MON 26 OCT	**Tutorial 2: patent text** 13:00–14:30 · 4P
MON 02 NOV	**Tutorial: knowledge spillovers** 13:00–14:30 · 4P
MON 09 NOV	**Tutorial: valuation algorithm** 13:00–14:30 · 4P
SUN 20 DEC	**Submission deadline** report, code, video
THU 07 JAN	**Final pitch session** 08:30–11:30 · room TBA

30 h **with me in total**
18 h of lectures + 12 h of Project Work
(6 × 90 min + 3 h pitch)

Tutor's lectures: Dr. Giacomo Masali

Two sessions with Dr. Giacomo Masali, each combining theory with a code tutorial.

MON
30 NOV

SESSION 1
17:30–19:00 · Room R1



Network theory with patent data

Patents cite each other and share inventors and technologies. Together they form networks we can build and analyse.

Theory **Code tutorial**

MON
14 DEC

SESSION 2
17:30–19:00 · Room R1



Innovation as recombination

New inventions often combine existing pieces of knowledge in new ways. How to see and measure this in patent data.

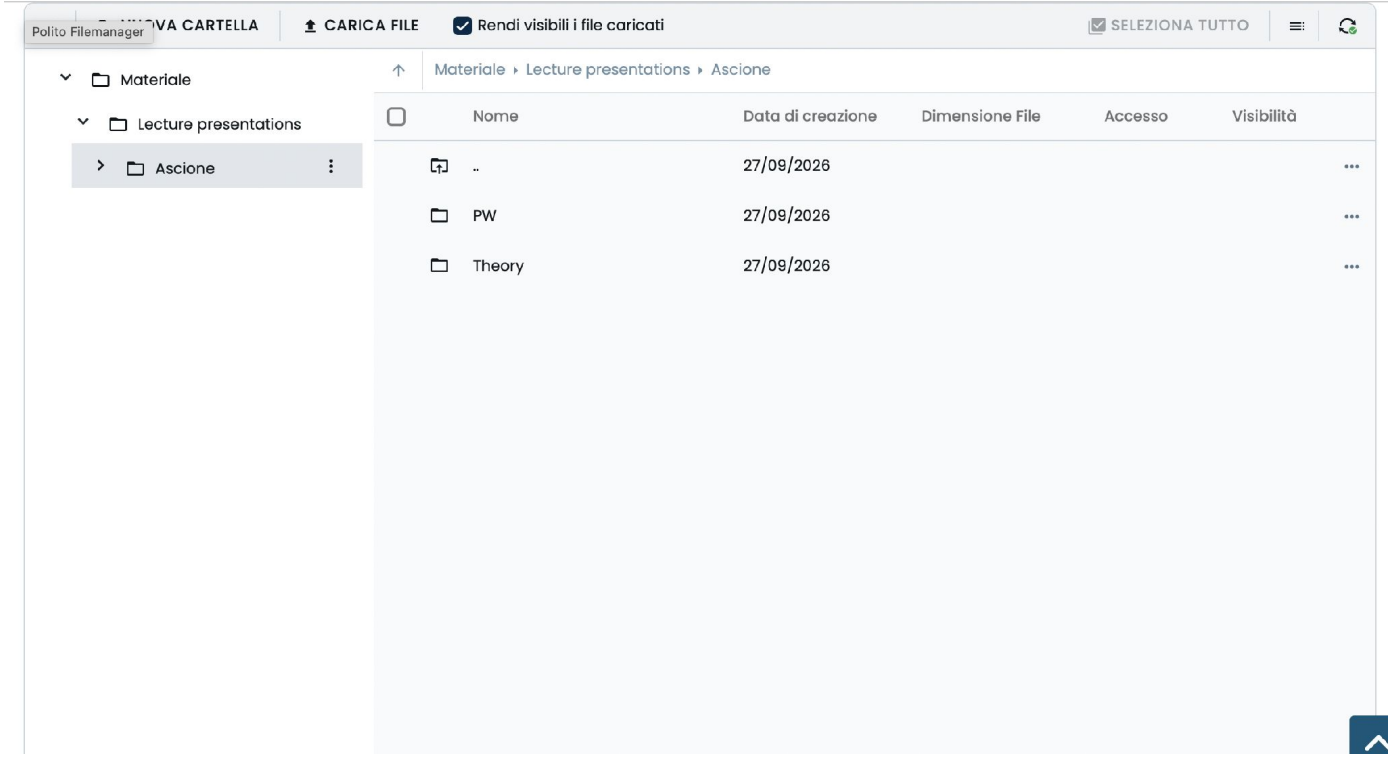
Theory **Code tutorial**



Same format for both: theory combined with a hands-on code tutorial.

Where to find the material

Moodle › Materiale › Lecture presentations › Ascione



Theory the slides of my lectures

PW everything for the Project Work sessions

WHAT YOU WILL FIND

Slides
in PDF or PPTX

Notebooks with code
sometimes homework, sometimes just code to play with

Run the notebooks anywhere, including the TIP platform we will use for Project Work data collection.

Evaluation

Project Work

Deliverables, criteria and deadline: at the **kick-off**, tomorrow.

Exam

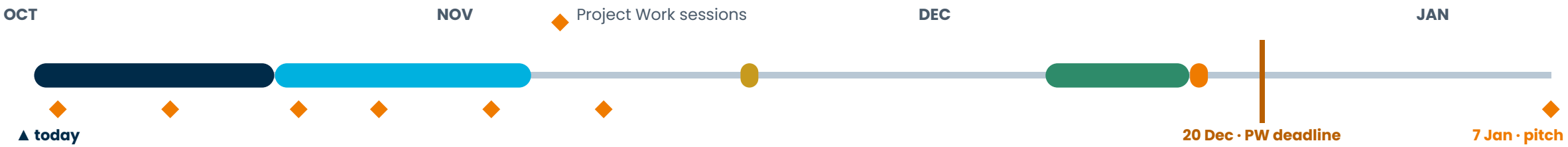
It combines two kinds of questions:

Theory
what the concepts are

Reasoning
applying them to a case

Anything that is not explained in class will ever be asked in the exam.

Lectures' flow



A IP foundations

How can innovators capture value from their ideas?

- Types of IPR and appropriability (today)
- Patents: filing, history, litigation
- IPR as a strategic choice

B Patents as data

How do we read technology through patent data?

- Patent intelligence and landscaping
- Knowledge spillovers
- Patent valuation

C IP and AI

What changes when AI creates and uses knowledge?

- IP and AI

D Digital strategy

How do digital firms compete and capture value?

- Digital strategy I, II, III
- AI case study

E Synthesis

How does strategy connect to the PW?

- Integrating strategy in Project Work

The goal of these lectures

Theoretical foundations + practical tools grounded in theory

Patent intelligence analyst

Apply what you learned during the class , and find a patent-related job

Where: Private consultancy (IP, strategy, investment) · European Commission · Start-ups

Research scientist

Get curious, invent more, do actual research around what what you learn

Where: thesis, PhD, conferences, publications · JRC (European Commission), OECD · Universities and research centres, patent offices (such as EPO or WIPO)

I share with you my competences: turn them into a job, or use them as the starting point to become a researcher.

Careers in IP: where to look

Three institutions that hire people with exactly the skills of this course.



EPO

European Patent Office · Munich, The Hague, Berlin, Vienna, Brussels

WHAT THEY DO

Examines and grants European patents for its 39 member states.

WHO THEY HIRE

Patent examiners with a STEM degree, data scientists, IT.

Careers page →



WIPO

World Intellectual Property Organization · Geneva

WHAT THEY DO

The UN agency for IP. Runs the international patent system (PCT) and publishes global IP data.

WHO THEY HIRE

Economists, data analysts, legal and IT profiles.

Careers page →



OECD

Organisation for Economic Co-operation and Development · Paris

WHAT THEY DO

Studies innovation, productivity and IP policy for its 38 member countries.

WHO THEY HIRE

Economists, statisticians, data scientists.

Careers page →

Why knowing a field matters more than ever

LLMs now help us write code. What they cannot easily replace is deep knowledge of the field the data come from.

Pure coding is assisted by LLMs

- Writing and debugging code
- Standard queries and data wrangling
- Boilerplate analyses and plots

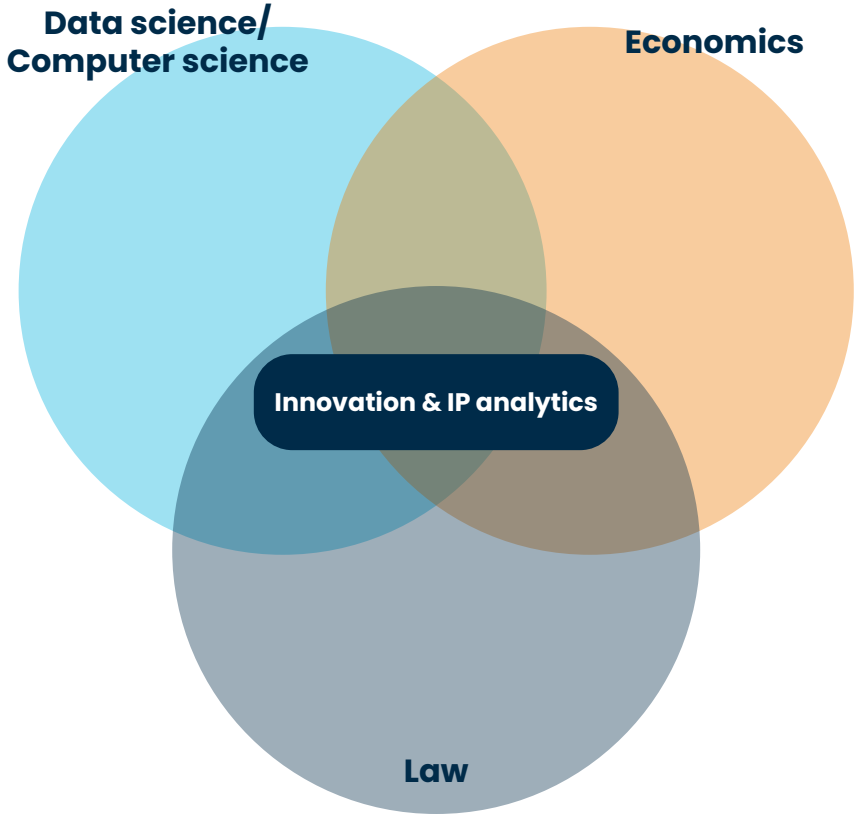
Hard to replicate: field expertise

- Knowing which question is worth asking
- Knowing what the data really measure, and their traps
- Judging whether a result makes sense
- Explaining it to a decision maker

Example: an LLM can write your query on patent data. It will not tell you that counting applications instead of patent families counts the same invention several times.

Interdisciplinary lectures

My research field sits at the intersection of three disciplines. This course draws on all three.



Data science (NLP + computer vision)

Creating algorithms and analysing large datasets: patent records, citation networks, the text of patents and scientific publications.

Economics

Why firms innovate and how they capture value: incentives, appropriability, knowledge spillovers, the value of patents, competition and strategy.

Law

IP is a topic of commercial law, and patents are legal texts: claims define what is protected, and courts decide validity and infringement.

Today's lecture

01 **Why intellectual property rights matter**

02 **Strategies and tools to protect innovation**

03 **A taxonomy of IPRs**

04 **An introduction to patents**

Quiz time at the end: bring your phone !

01

Why intellectual property rights matter

Intangible capital and the appropriability problem

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Beyond tangible capital

Understanding modern firms, and modern economies, requires broadening the concept of capital beyond tangible assets.

Tangible assets

WHAT

Physical resources: buildings, machines, vehicles, IT equipment.

ROLE

Provide the means of production and support day-to-day operations.

R&D and intangible assets

WHAT

Knowledge capital: R&D investments, patents, proprietary technology, software, data, know-how.

ROLE

Drive innovation, differentiation and future growth options.

How important are tangible and intangible assets in business today?

What is a firm really worth?

Ferrari must lose one of these two things overnight. Which one hurts more?

A



Every factory burns down

or

B



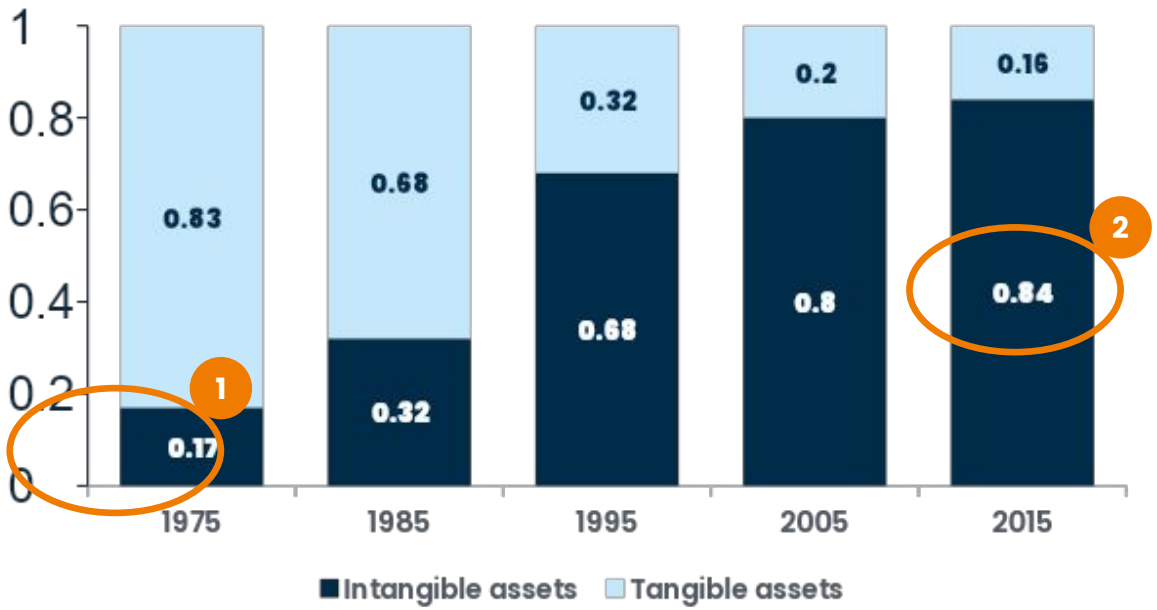
It loses its name, logo and know-how

Factories can be rebuilt. A name and know-how cannot.

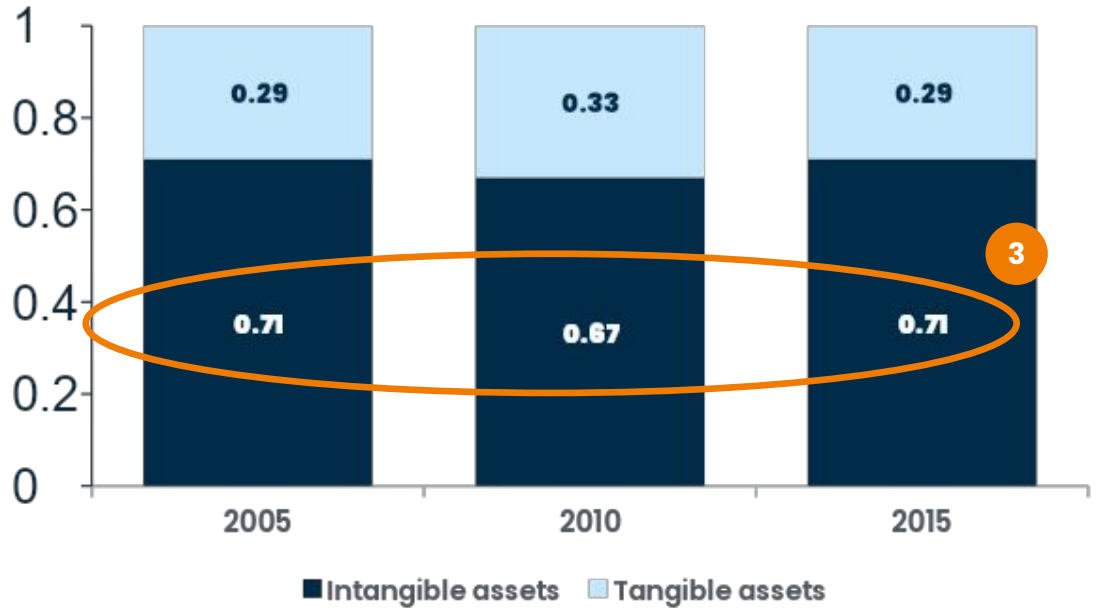
Intangible assets now dominate firm value

Share of market value from intangible vs tangible assets (Ocean Tomo, 2022)

UNITED STATES · S&P 500



EUROPE



In the EU, IP-intensive industries **generate 45% of GDP** and **63 million jobs (29%)**, paying wages 47% higher than other sectors.

The core problem

"If I am the CEO of a tech start-up and I spend time and money to invent something, ***how can I prevent other people from copying my idea and benefiting from it?"***

I invest

Time and money go into R&D to create something new.

›

They copy

Competitors imitate without investing, so they can sell the product much cheaper.

›

I exit

My start-up cannot recover its investment and has to leave the market.

So... what can be done to avoid this situation?

Knowledge as a public good: the classic matrix

Economists classify goods along two dimensions: rivalry and excludability.

	RIVAL	NON-RIVAL
EXCLUDABLE	**Private goods** **Rival and excludable.** If I use it, you cannot, and the owner can make you pay for it.	**Club goods** **Non-rival but excludable.** Many can use it at once, but only those who pay get access. **Patented or secret knowledge**
NON-EXCLUDABLE	**Common-pool goods** **Rival but non-excludable.** Anyone can access it, but every use leaves less for the others.	**Public goods** **Non-rival, non-excludable.** My use does not reduce yours, and no one can be kept out. **Knowledge, in principle**

▲ **IPRs and secrecy add excludability**

TWO CHARACTERISTICS

Rivalry

If I use it, can you still use it? A sandwich is rival; a theorem is not.

Excludability

Can the owner stop those who do not pay from using it?

Classify these goods

Which quadrant does each one belong to? Call out the numbers.

	RIVAL	NON-RIVAL
EXCLUDABLE	**Private goods** ?	**Club goods** ?
NON-EXCLUDABLE	**Common-pool goods** ?	**Public goods** ?

FIVE GOODS TO CLASSIFY



1 The last slice of pizza at a party



2 Your Netflix password, shared with five flatmates



3 The San Giovanni fireworks over the Po



4 An espresso at the bar



5 Pythagoras' theorem

Solution: where does each good go?

Excludability is often a matter of enforcement, not a fixed property of the good.

	RIVAL	NON-RIVAL
EXCLUDABLE	**Private goods**  4 An espresso at the bar	**Club goods**  2 The shared Netflix password
NON-EXCLUDABLE	**Common-pool goods**  1 The last slice of pizza	**Public goods**  3 San Giovanni fireworks  5 Pythagoras' theorem

Why knowledge behaves like a public good

1 Non-rival

Using an idea does not use it up. Once produced, knowledge can be reused by anyone at almost zero marginal cost.

Training a large model costs millions; copying its weights costs almost nothing.

2 Hard to exclude

Once disclosed, ideas travel: through publications, reverse engineering, and people changing jobs.

A published algorithm can be re-implemented in an afternoon.

3 …but only quasi-public

Using knowledge requires skills. **Tacit knowledge and absorptive capacity** (Cohen & Levinthal, 1990) limit who can actually use it.

Reading a paper is not the same as being able to replicate its results.

Consequence: markets alone under-invest in producing knowledge (Nelson, 1959; Arrow, 1962).

The appropriability problem

DEFINITION

Appropriability: the ability of the innovator to reap the economic benefits of its investments in innovation (R&D, learning…) by limiting imitation by rivals.

WHY IT IS A PROBLEM

- **Knowledge is a (quasi-)public good:** once out, it is hard to exclude others from using it (Arrow, 1962)
- **Free riders** copy the innovation without bearing the cost of R&D
- **Result:** private under-investment in R&D and under-provision of innovation

1 The innovator invests in R&D

↓

2 Knowledge is (quasi-)public

↓

3 Free riders copy

↓

4 The innovator cannot fully capture the returns

↓

5 Less R&D, less innovation than the social optimum

The appropriability problem: solutions and incentives

1 Government ("public") intervention

- **Public R&D expenditure** (**Samuelson** approach): funding universities and public research organisations
- **Subsidies and fiscal incentives** (**Pigou** approach): R&D grants and tax credits to private firms
- **Public procurement:** the government, as sole buyer, commits to buy only from the innovator (e.g. health, defence)
- **Prizes:** e.g. the DARPA Grand Challenge

2 "Private" mechanisms

Their effectiveness differs across sectors.

FORMAL · LINDAHL APPROACH

Intellectual property rights (IPRs)

INFORMAL · STRATEGIC

Trade secrets, lead time, complementary assets

02

Strategies and tools to protect innovation

Informal methods, formal methods, and the trade-off between them



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Informal methods

Strategic methods to keep a competitive advantage over imitators without any direct legal protection.

Secrecy

Keep knowledge inside the firm: trade secrets, confidential information, non-disclosure agreements and non-compete clauses.

e.g. **Coca-Cola recipe**



Complexity

Make knowledge so specific and complex that it is hard to imitate, e.g. through reverse engineering.

e.g. **Intel microchips**



Lead time

Exploit first-mover advantages (natural imitation delays, reputation) and keep introducing incremental innovations.

e.g. **Tesla Model S**



Complementary assets

Distribution, after-sales services, marketing and advertising that imitators lack.

e.g. **Apple ecosystem**

Why firms often prefer informal methods

USUALLY PREFERRED BECAUSE THEY ARE



Less costly



Less time-consuming

...BUT THERE ARE IMPORTANT EXCEPTIONS



Sector matters

In pharma and chemistry, knowledge tends to be codified: spillovers are large and imitation is easy and cheap, so informal methods protect less.



Size matters

Large firms use formal methods more than small firms, because they can afford the costs (filing and renewal fees).

Formal methods: intellectual property rights

Intellectual property rights are exclusive rights granted to innovators over the use and economic exploitation of their ideas or creations. Formally, they create **temporary monopolies** on a given idea, invention or technology.



You create something new

An invention, a design, a song, a piece of code.

→



You get exclusivity

For a limited time, others cannot copy or sell it without your permission.

→



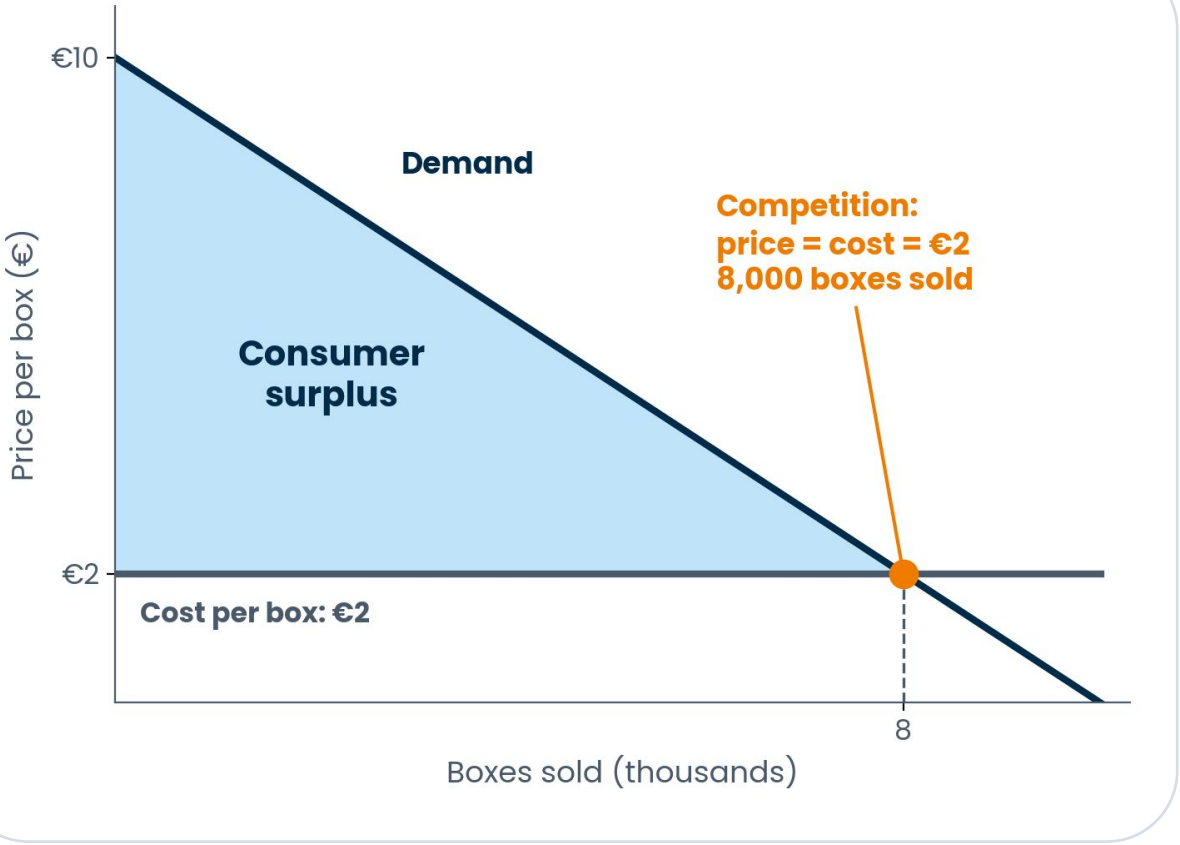
The right expires

Anyone can then use it for free. Trademarks are the exception: they can be renewed forever.

Which right, and for how long? It depends on what you created. We see each type in section 03.

Patents generate a (microeconomic) problem

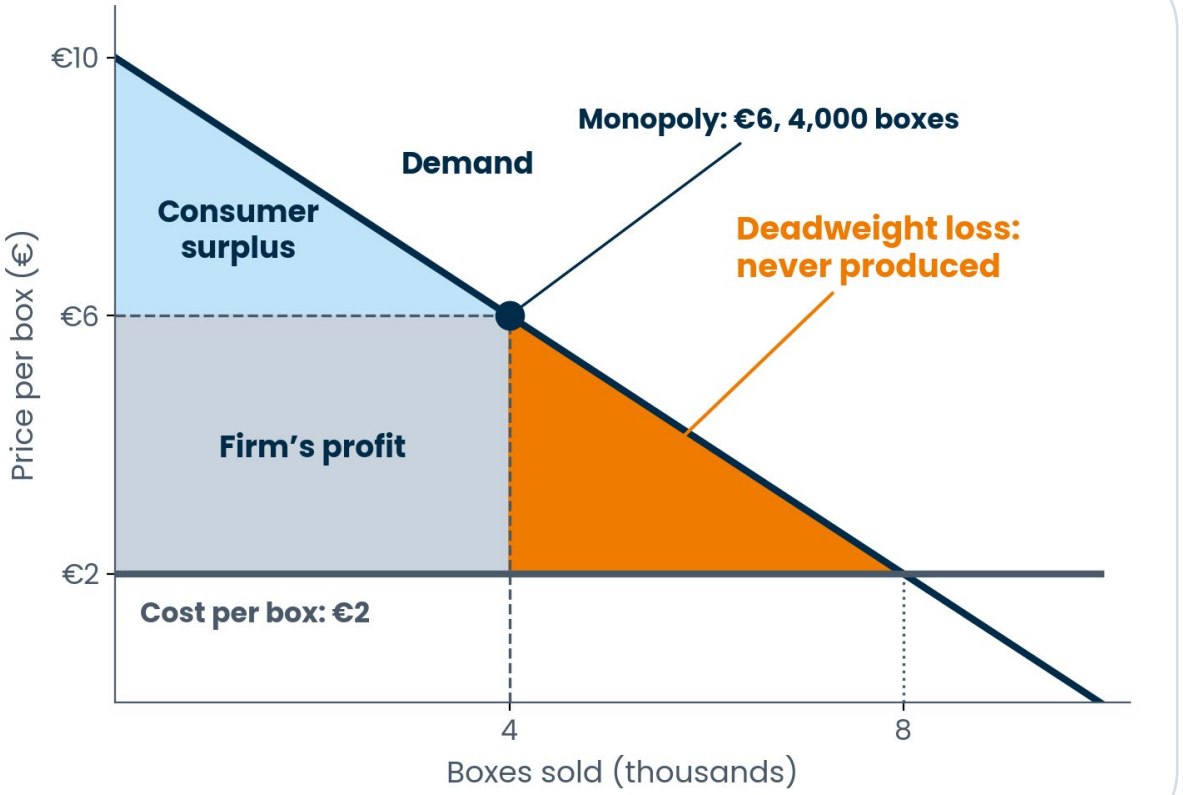
Example: a medicine that costs €2 per box to make. Many firms are able to produce it.



Perfect competition	Many firms sell the same product. Charge more than the others and customers walk away.
Demand curve	Each point is a buyer and the most they would pay. Lower price, more buyers.
Cost per box	What it costs to make one more box: €2. Nobody can sell below it for long.
Consumer surplus	What buyers would pay minus what they actually pay: their "good deal" (blue area).

Result: price falls to cost. 8,000 boxes sold at €2.

Now one firm holds the patent. It is the **only seller (temporary monopoly)**, so it picks the price.

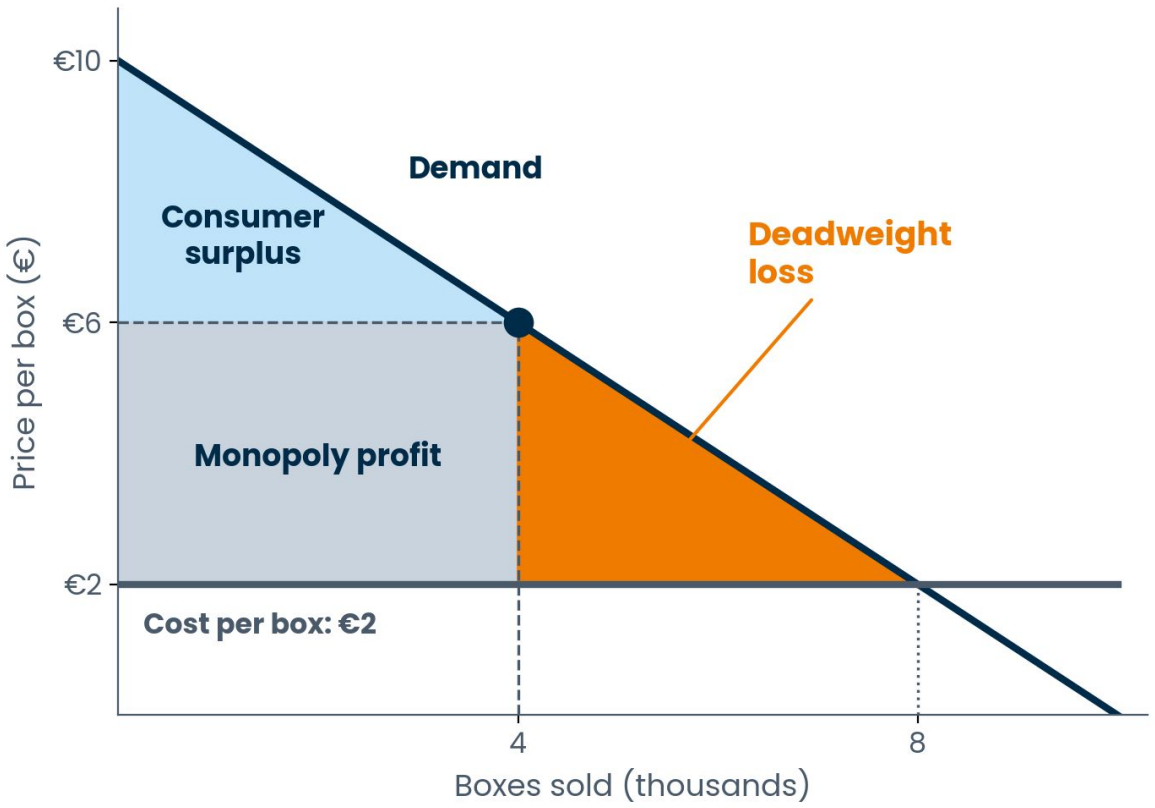


HOW THE MONOPOLIST PICKS THE PRICE

Price	Boxes sold	Profit = (price − €2) × boxes
€10	0	€0
€8	2,000	€12,000
€6	**4,000**	**€16,000**
€4	6,000	€12,000
€2	8,000	€0

	Competition	Monopoly
Price	€2	**€6**
Boxes sold	8,000	**4,000**
Consumer surplus	€32,000	**€8,000**
Deadweight loss	**€0**	**€8,000**

The trade-off: static vs dynamic efficiency



THE PROBLEM

With a monopoly, even a temporary one, consumer surplus is lower than under competition, and a **deadweight loss** appears.

Static efficiency
Lower prices today

Dynamic efficiency
More innovation tomorrow

Example: a vaccine without a patent costs less today, but in the long run there are fewer incentives to innovate. Policy debate: finding the right balance.

Two families of IPRs

Intellectual property rights

Copyright

Rights of authors of **literary** and **artistic** works.

- **Books**
- **Music**
- **Films**
- **Software**

Industrial property

Rights with a direct **industrial application**.

- **Distinctive signs** — **Trademarks**
- **Inventions** — **Patents, designs, utility models**

03

A taxonomy of IPRs

Copyright, trademarks, designs, utility models, and how they combine. Patents are IPRs too.



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Copyright

Protection for **artistic and creative works**: books, songs, films, **software**.

What it protects

The expression of an idea, not its content:

Registration

Not needed: the right arises automatically with the act of creation (registration is required to bring a lawsuit in some countries, e.g. the US).

Cost and coverage

Free of charge, in all countries that recognise copyright by law (signatories of the Berne Convention, 1886).

Duration

70 years after the author's death, with no annuities. US anonymous or team works: 95 years from publication or 120 from creation, whichever expires first.

Exceptions

Moral rights, and "fair use" / "fair dealing": parody, private study, research and education, archives, reporting.

Copyright: content vs expression



Books of Magic (1990)

vs.



Harry Potter (1998)

Example

A young boy with glasses discovers he has magical powers.

Similar ideas, eight years apart. Is this a copyright infringement?

Copyright protects the **expression**, not the **content** of an idea.

Fair use: copying without asking

Copyright has exceptions. The US uses a flexible test ("fair use"); the UK and the EU list specific cases such as quotation, **parody** and teaching ("fair dealing").



Google Books

US, 2015

Google scanned millions of books so you can search inside them, but shows only short snippets.

✅ **Fair use**

New purpose: a search tool, not a substitute for the book.



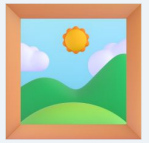
"Pretty Woman" parody

US Supreme Court, 1994

2 Live Crew turned Roy Orbison's love song into a rap parody that mocked it.

✅ **Fair use**

A parody comments on the original, so it can be fair use.



Warhol's Prince

US Supreme Court, 2023

A Warhol print based on a photographer's portrait of Prince was licensed to a magazine cover.

❌ **Not fair use**

Same use as the photo, a magazine image, so it competes with it.

Is training an AI on books fair use?
Courts are deciding now.

SUPREME COURT

Supreme Court rules against Warhol foundation in copyright fight over Prince images

The dispute centered on whether artist Andy Warhol's silk-screen prints of Prince based on a photo taken by Lynn Goldsmith violated copyright law.



Trademarks

A trademark can be any sign (word, logo, shape, picture) that distinguishes the goods and services of one trader from those of another.

Rationale	A trademark signals consistent quality: it lowers consumers' search costs, so the firm can charge a higher price.
Duration	Unlike patents, **indefinite**, as long as the firm keeps trading in the covered classes. Initial 10-year period, then renewal fees.
Coverage	As for patents: limited to the countries of the offices and the product classes where it is registered.



Shapes can be trademarks too
The Coca-Cola bottle and the Toblerone bar



i'm lovin' it®

Designs and utility models

1 Design rights

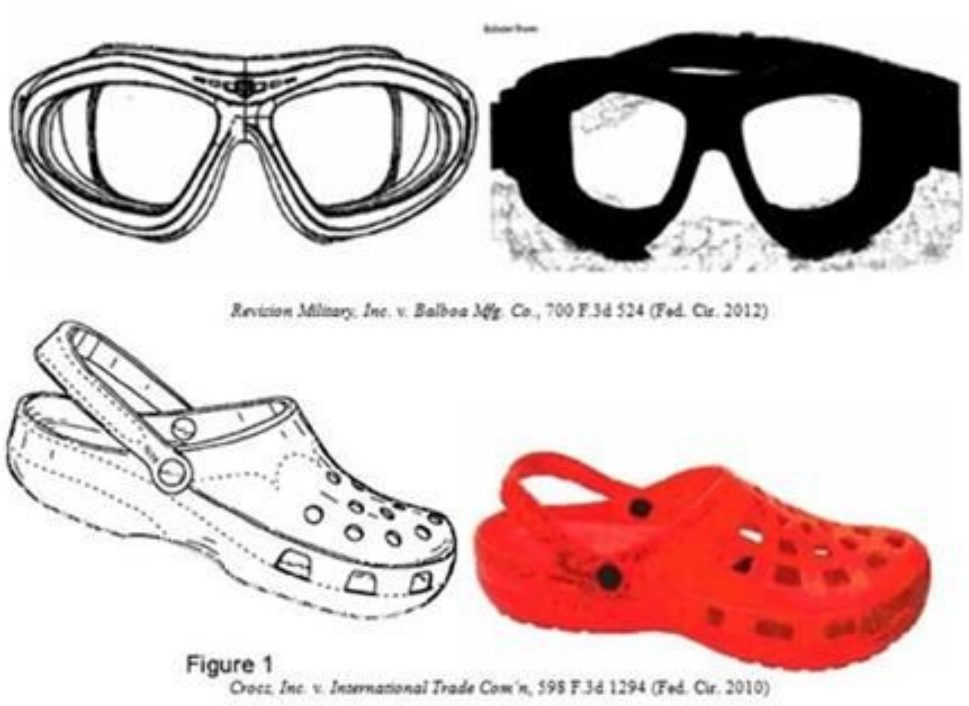
Protect the external, visible features of a product.

- **Scope:** can include items where **design is integral to performance**, e.g. semiconductor chip layouts
- **Duration:** variable, around 15 years

2 Utility models

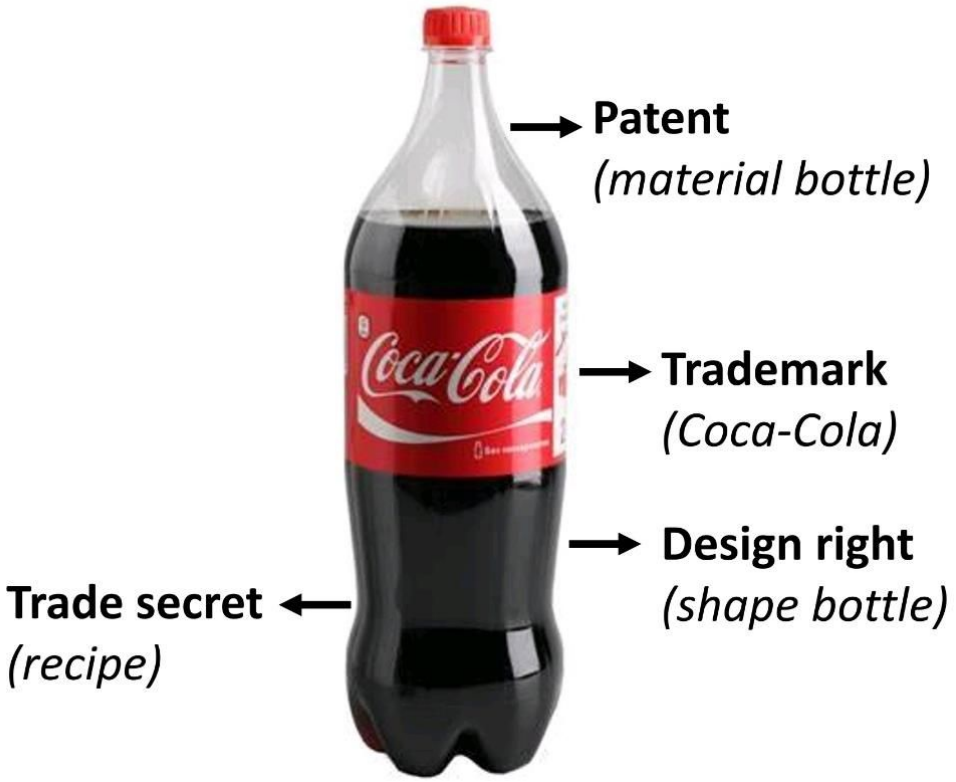
A blend of a design and a patent.

- **Examination:** usually limited
- **Duration:** short, often 6 to 15 years



Design disputes: Revision Military v. Balboa (2012); Crocs v. ITC (2010)

Combining IPRs on a single product



One product, four layers of protection

Different rights protect different aspects of the same product, and firms combine formal and informal methods.

Patent	→ the material of the bottle
Trademark	→ the Coca-Cola name
Design right	→ the shape of the bottle
Trade secret	→ the recipe

Overview of IPRs

RIGHT	WHAT IT PROTECTS	HOW YOU GET IT	HOW LONG	EXAMPLE
 **Patent**	New technical inventions with industrial application	Application and examination	**Up to 20 years from filing**	A new battery chemistry
 **Copyright**	Original creative works, software included	Automatic, when you create it	**Author's life + 70 years**	Your code, a song, this deck
 **Trademark**	Signs that identify a brand	Use and/or registration	**10 years, renewable forever**	A brand name or logo
 **Design**	The external look of a product	Registration	**EU: up to 25 years**	The shape of a chair

04

An introduction to patents

What a patent is, the rights it confers, and what can be patented

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What is a patent?

A contract between an inventor and the state

THE INVENTOR GETS

Protection for about 20 years: the right to exclude others from making, using, selling and importing the invention

- to recoup the investment in R&D
- to strengthen market position and competitiveness

In exchange: the obligation to disclose the invention



THE STATE GETS

Publication of the invention

- to spread new technical knowledge
- to avoid duplication of R&D
- to foster further innovation
- to improve tradability of technology

Rights conferred by a patent

Territorial

A patent in one state does not affect patents in other states.

Timed

Protection runs from a certain date to another date.

Transferable

The patent can be assigned or sold, or licensed to others.

Exclusion

Nobody else can produce, use, sell or import the invention.

ATTENTION

A patent is a right to exclude, not necessarily a right to exploit. Using your own invention may still depend on others' rights: if another company owns a molecule essential to your formula, you need a licence.

What can and cannot be patented

For an invention to be patentable, it must meet three requirements.

Novelty

New to the world: not available to the public anywhere before the filing date.

Inventive step

Not an obvious solution for an expert in the field.

Industrial application

Concrete or potential industrial use: a theorem or formula cannot be patented per se.

LIMITS, AND ASYMMETRIES ACROSS COUNTRIES

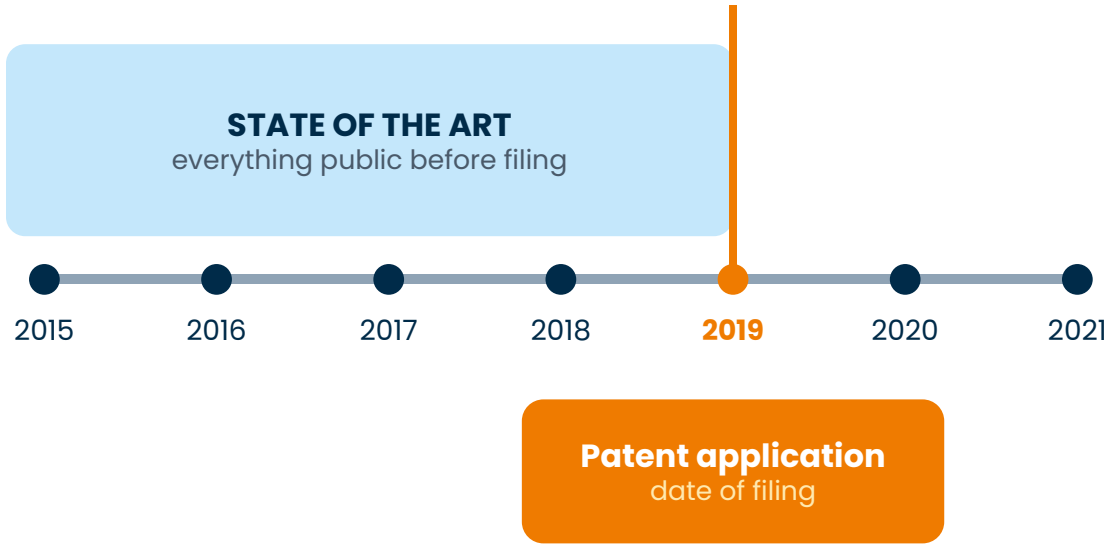
- Methods of treatment, diagnosis and surgery on the human or animal body
- Plant and animal varieties
- Software and algorithms: treated differently in Europe and in the US

Novelty: when is an invention new?

An invention is new when it is not part of the state of the art (prior art).

- **Prior art:** all knowledge disclosed and accessible to the public, worldwide, before the filing date
- **In the EU:** everything made available to the public before the date of filing, in any form
- **Examples:** earlier patents, scientific publications, products on the market, public talks

THE FILING DATE DRAWS THE LINE



Inventive step: when is an invention inventive?

When it is not obvious to the person skilled in the art, in view of the state of the art.

"[The] non-obviousness principle asks whether the invention is an adequate distance beyond or above the state of the art."

- **It measures** the inventive height of a patent
- **It is the most debated** and ambiguous requirement: it often splits patent offices, examiners, lawyers and courts



The person skilled in the art

- is a skilled practitioner in the relevant technical field
- has access to the entire state of the art
- knows the general technical knowledge
- is capable of routine work

Knows everything, but has zero imagination!

Exercise: could Homer Simpson’s chair be patented?



Novelty **YES**

Inventive step **NO**

Obvious for someone working in the field.

Industrial application **?**

To be defined.

Before moving on: rationale and concerns

Why the patent system exists

- Incentives for private investment in R&D
- Incentives to disclose information
- Dynamic efficiency, at the cost of static welfare losses

...but it raises ethical concerns

- Excessive concentration of monopoly power
- Socially sub-optimal allocation of resources

Keep this tension in mind: it comes back with patent litigation, IP strategy, and IP and AI.

A real case: patents vs access to AIDS medicines

South Africa, 1997–2001. Millions of people with HIV, and medicines most of them could not afford.

WHAT HAPPENED



1997 South Africa passes a law allowing cheaper imports of patented medicines.



1998 39 pharmaceutical companies sue the government to block it.



2001 After worldwide public protests, the companies withdraw the lawsuit.



2001 Doha Declaration: WTO members agree patents should not stop countries from protecting public health.



One year of HIV treatment, per patient

$10,000+

patented brand-name drugs, around 2000

$350

generic version offered by Cipla (India), 2001

It came back with COVID-19: India and South Africa asked the WTO to waive vaccine patents (2020). Who pays for the next drug?

QUIZ TIME

Test what you learned today

Scan the QR code with your phone and answer the questions.



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Questions?

Tomorrow
Project Work kick-off · Tuesday 6 October, 17:30–19:00 · Room R3

Next lecture
Patents: filing, history and litigation · Monday 12 October, 13:00–14:30 · Room 4P

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